Input\nCompleted genome analysis job (job_id)\n+ patient biomarker measurements\n(timestamp, biomarker name, value, unit)
Genetic Profile\nSource
Genome job complete
No job linked
Derive from Job Results\nprofile_from_job.py\nExtract pharmacogenomically relevant\nvariants from pipeline output
Synthetic Profile\ngenerate_synthetic_profile()\nSample genotypes under Hardy-Weinberg\nequilibrium for 25 catalog SNPs
SNP Effect Catalog
(engine/tracking/genetics.py)\n25
NOS3, VEGFA, COL1A1, MMP1\nVascular repair, angiogenesis, collagen synthesis\n(BPC-157, TB-500, GHK-Cu targets)
IL6, TNF, IL10, TLR9, VDR\nInflammation and immune axis\n(KPV, Thymosin Alpha-1, LL-37 targets)
IGFBP3, IGF1, GHR\nGrowth hormone / IGF axis\n(CJC-1295, Ipamorelin, GHRP-2 targets)
PPARG, TCF7L2, FTO, MC4R\nMetabolic and insulin axis\n(MOTS-c, AOD-9604, Melanotan II targets)
BDNF, ANKK1, MTHFR, CLOCK\nNeuro-cognitive and circadian\n(Semax, Selank, Dihexa, Epitalon targets)
effect weights\nfor prior
Prior Derivation (genetics.py
_prior)
Aggregate Variant Weight\ng = sum of dosage x effect_weight\nper peptide across all relevant SNPs\ndosage: hom_ref=0, het=1, hom_alt=2
Responder Strength Prior\nr ~ Normal(1 + 0.72 x tanh(g), sigma_r)\nsigma_r = max(0.06, 0.18 / sqrt(1 + 0.5n))\nr=1.0 is average responder, r=0 is non-responder
Biomarker Prior on theta\ntheta = fractional change at panel timeframe\nmu_prior = r_mean x expected_pct_panel\nsigma_prior = hypot(sigma_r x |expected_pct|,\n 0.20 x |expected_pct|)\nfloor: sigma >= 0.04
Likelihood Computation from
ements (bayes.py)
Exponential Approach Factor\na(w) = 1 - exp(-w / tau)\ntau = panel time constant (weeks)\na(w) represents fraction of asymptotic\nresponse achieved at week w
Joint Bayesian Linear Regression\njoint_fit_likelihood()\nModel: y_i = b + phi x a(w_i) + epsilon_i\nwhere phi = b x theta\nPrior on b: Normal(baseline_panel, sigma_b)\nWeak prior on phi to prevent singularity\nIterates up to 4 rounds refining sigma^2\nfrom residuals when n >= 3
Delta Method Recovery\ntheta_hat = phi_hat / b_hat\nVar(theta) via delta method:\nVar(phi)/b^2 + Var(b) x phi^2/b^4\n- 2 x Cov(b,phi) x phi/b^3\nStudent-t inflation correction:\nsd_theta x= t_0.975(df) / z_0.975\n(prevents under-coverage at small n)
Fallback: Inverse-Variance Pooling\neffective_observation()\ntheta_hat_i = (y_i/baseline - 1) / a(w_i)\nsigma_i = noise_pct / a(w_i)\nPooled: tau_total = sum(1/sigma_i^2)\nmu_pool = sum(theta_hat_i / sigma_i^2) / tau_total
Normal-Normal Conjugate
Update (bayes.py
update)
Precision Combination\ntau_prior = 1 / sigma_prior^2\ntau_like = 1 / sigma_like^2\ntau_post = tau_prior + tau_like
Posterior Mean\nmu_post = (tau_prior x mu_prior\n + tau_like x mu_like) / tau_post\n(weighted average toward data as n grows)
Posterior SD and Credible Interval\nsigma_post = sqrt(1 / tau_post)\n95% CI: mu_post +/- 1.96 x sigma_post\ncredible = True if CI excludes zero
Posterior Predictive Trajectory
(bayes.py predictive_curve)
Mean Trajectory\nV(w) = baseline x (1 + mu_post x a(w))\nProjected over a fine week grid
95% Credible Band\nVar(V) = sigma_b^2 x (1 + mu_theta x a)^2\n + baseline^2 x a^2 x sigma_theta^2\nBand = mean +/- 1.96 x sqrt(Var(V))\nBaseline uncertainty propagated when\nno pre-treatment sample is available
Expected Change Window
(bayes.py expected_window)
weeks_min (detection threshold crossing)\nSolve: |theta_post| x a(w) = detection_noise_pct\nweeks_min = -tau x ln(1 - detection_pct / |theta_post|)
weeks_max (plateau)\nweeks_max = -tau x ln(1 - 0.95) (~3 x tau)
Direction\nincrease | decrease | inconclusive\nInconclusive when 95% CI includes zero\nor |theta_post| <= detection threshold
Output per (patient, peptide, biomarker)\nPosterior: mu_post, sigma_post, 95% CI, n_effective\nPredictive curve: mean + credible band over week grid\nExpected window: weeks_min, weeks_max, direction\nAll stored in PostgreSQL / SQLite tracking database
read history\nwrite posterior
Tracking Database\nSQLite / PostgreSQL\nPatients, measurements\nposteriors, cohort data